

Summary of UAV Connection and UAV Delivery

1. Aerial Platform Capabilities for Wireless Services

- Drone Characteristics
 - Pros: wide-area coverage, high mobility, easy to maintain, no fuel required (environment friendly), ad-hoc network friendly, widely available
 - Cons: most time low payload (depend on the size), cannot deploy and operate under several different types of weather conditions, bad duration (in 1 hour)
- Aircraft Characteristics
 - Pros: significant payload capability, extreme duration (over 3 days), optimum altitude, wide area coverage, moving coverage, high mobility, can be deployed and work under multiple complex weather conditions
 - Cons: hard to maintain, not ad-hoc network friendly, not air traffic friendly, not easy to handle and deploy
- Airship Characteristics (aerostatic platforms)
 - Pros: high payload, wide area coverage, moving coverage, ad-hoc network friendly, optimum altitude (up to 30 km), high duration (several years) without moving
 - Cons: low mobility (large volume), requiring high-strain envelope material, need ground operations center, hard to maintain, not air traffic friendly
- Helikite platform: a combination of Helium balloon and kite. Using both wind lift and helium lift for high altitude flying in multiple weather conditions, can carry more payload than other aerostat, no need electric for flying and operating, only need electric for sensors working, have very few legal obstacles compared to other UAV platforms.

Compared with Aerial Base Station (ABS) and Ground Base Station (GBS)

- ABS: An Aerial base station (ABS), also known as unmanned aerial vehicle (UAV)-mounted base station (BS), is a flying antenna system that works as a hub between the backhaul network and the access network.
 - Pros: quick deployment, high flexibility, wide line-of-sight (LoS), provide extra communication capacity to reduce the burden of GBS
 - Cons: low durability, weather conditions, high cost (deployment and maintenance) signal jamming, legal limitations
- GBS
 - Pros: stable. It can provide continuous and reliable communication services. Power supply, high capacity, with large processing ability and high bandwidth for large number of users at the same time, suitable for city and high stream regions. Complete infrastructure (optical fiber, power station), less legal limitations.
 - Cons: low deployment flexibility (need to build infrastructure first), limited LoS (obstacles like buildings and mountains), limited coverage (may exists signal dead zone because of buildings, mountains and far regions), cannot handle emergency (damage of ground infrastructure)
- Best solution: combination of ABS and GBS. Using ABS to fix the dead zone of GBS and using GBS to enhance the signal power for FANET

Secure Drone Communication

- The drones communicating via cellular data use radio signals to communicate with the controller. The controller sends the radio signals through its transmitter and these are received by the drone through its receiver.
- Bluetooth vs Wi-Fi communication
 - Bluetooth: used for short-range device-to-device communication. Short range (up to 30 feet), data transfer rate up to 2 Mbps, low power consumption, less secure and less expensive
 - Wi-Fi: used for Local area networking and internet access. Long range (several hundred feet), data transfer rate up to 1 Gbps, high power

consumption, more secure and more expensive.

- The better way to communicate between drones is to use Bluetooth because of the low power consumption. The durability of UAVs is important. UAVs need to fly closer to each other for data transforming because of the short connection range.
- The better way to build the system of ABS and GBS is to use Wi-Fi. Wi-Fi has the long-range connection range so that the base station of aerial and ground does not need to be dense. Meanwhile, a base station using Wi-Fi communication can have a wider range of signal coverage and will have less dead zone. High data transfer rate can make the base station have more efficiency data interaction with the warehouse to deal with a large amount of data.
- Blockchain based UAV security communication
 - A blockchain is a distributed ledger with growing lists of records (blocks) that are securely linked together via cryptographic hashes. A block in the blockchain contains an index, a timestamp of message recorded, the data itself, a previous hash of previous block that connect to this block, a hash of this block.
 - Application in network of UAVs: detecting poisoned data and insuring data integrity, securing UAV data types, avoiding mid-air collision of UAVs, securing data dissemination to end users, securing drone communications to avoid attacks such as man-in-the-middle attack
 - The communication of ABS and GBS can apply blockchain based communication network for data transfer between warehouse and base stations. Each swarm UAV can become a node of blockchain network and help to keep a copy of block information to avoid malicious tampering and can be used to verify that the message is authorized or not.